Entropy cannot tell electric tides from magnetic ones. Area can.

An exact second-order mismatch for free gravitons in a flat ball

Andreas Houg

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Abstract

Take a ball in flat space and put a weak tidal gravitational field through it. Two quantities respond at second order in the field: the relative entropy of the gravitons inside the ball, and the area of its surface. The first law of causal diamonds, carried to second order, would make them equal. We compute both exactly in free linearised gravity. On a static electric tidal field of multipole order l they agree. On a stationary magnetic one the entropy is only $(l+1)/(l+2)$ of the area term. For a uniform field in a ball of radius R the entropy is the Bel–Robinson super-energy,

$$S = \frac{\pi R^6}{420\,G}\big(\mathcal{E}_{ij}\mathcal{E}_{ij} + \mathcal{B}_{ij}\mathcal{B}_{ij}\big),$$

while the area term is $\frac{\pi R^6}{420\,G}\,\mathcal{E}_{ij}\mathcal{E}_{ij}$ for an electric field and $\frac{4}{3}\cdot\frac{\pi R^6}{420\,G}\,\mathcal{B}_{ij}\mathcal{B}_{ij}$ for a magnetic one. The area terms are in print (Jacobson, Senovilla and Speranza, 2017); the entropy is new. The entropy is blind to electric–magnetic duality and the area is not. No local scalar removes the mismatch, and it vanishes identically when the ball's conformal Killing vector is replaced by a Killing vector.

1 The result

Units: $\hbar = c = 1$, Newton's constant G kept. Let B be the unit ball in the slice $t = 0$ of Minkowski space and D its causal diamond. Let the spacetime carry a weak vacuum tidal field of amplitude ε, with electric and magnetic parts

$$\mathcal{E}_{ij} = R_{0i0j}, \qquad \mathcal{B}_{ij} = \tfrac{1}{2}\,\epsilon_{ikl}\,R_{0j}{}^{kl}.$$

We compare two numbers, both of order ε^2.

- **The entropy side.** Quantise the linearised field. A tidal field is a coherent state of gravitons (only its data near the ball matter; take any finite-norm solution that agrees with the tidal field there). S is its relative entropy to the vacuum, both restricted to the observables in D. It measures how well an observer confined to the diamond can tell the tidal field from empty space. For a coherent state the von Neumann entropy of the ball does not change, so S is the change in modular energy.
- **The area side.** Let $F = \mathrm{Area}(\mathcal{S}) - 2\,\mathrm{Vol}(\Sigma)$ for a closed surface $\mathcal{S}$ near the unit sphere and a slice Σ it bounds; the 2 is the mean curvature of the unit sphere. The unit ball makes F stationary in flat space under all variations of the surface and the slice. The area side is $-\Delta F/4G$, the drop of the stationary value of F in the perturbed spacetime.

For conformal matter the two are equal at first order: this is the first law of causal diamonds [1], and it is the content of Jacobson's derivation of the Einstein equation from entanglement [2]. For non-conformal matter they already differ, by a local scalar (Section 5). Gravitons first appear at second order in the metric perturbation, so for them the comparison is a second-order one.

Write $\psi = x_i x_j \mathcal{E}_{ij}$ or $x_i x_j \mathcal{B}_{ij}$ for the tidal potential of a multipole of degree $l \geq 2$ (a harmonic polynomial of degree l), and $c_l = 4\,(l-2)!/(l+2)!$.

Result. *In free linearised gravity on the flat unit ball, for a tidal multipole of degree $l \geq 2$:*

$$8G\,S = c_l\,\frac{l+1}{2}\int_B \big(\psi_{\mathcal{E}}^2 + \psi_{\mathcal{B}}^2\big) \quad \textit{(any static electric plus stationary magnetic data)}, \tag{1}$$

$$8G\,(\textit{area side}) = c_l\,\frac{l+1}{2}\int_B \psi_{\mathcal{E}}^2 \quad \textit{(static electric data)}, \tag{2}$$

$$8G\,(\textit{area side}) = c_l\,\frac{l+2}{2}\int_B \psi_{\mathcal{B}}^2 \quad \textit{(stationary magnetic data)}. \tag{3}$$

So at the stationary surface the first law holds on electric data and fails on magnetic data, where

$$\frac{\textit{entropy}}{\textit{area side}} = \frac{l+1}{l+2} = \frac{3}{4},\ \frac{4}{5},\ \frac{5}{6},\ \cdots$$

How far each line reaches:

- (1) is exact to all orders, because the state is coherent and the theory is free. It has no cross term between the sectors (it would be odd under parity).
- (3) is proved for every l.
- (2) is computed exactly for $l \leq 12$, and for every l given the closed form of Section 4.
- When both sectors are present, the area side is the sum of (2) and (3) only if the ball is not pushed along a boost. For $l = 2$ that requires $\mathcal{E}$ and $\mathcal{B}$ to share principal axes, $\epsilon_{kil}\mathcal{E}_{ij}\mathcal{B}_{lj} = 0$. This is argued from the symmetry of the zero modes; the general mixed case is not computed.

For $l = 2$ the fields are uniform and $\int_B \psi^2 = \frac{8\pi}{105}\mathcal{E}_{ij}\mathcal{E}_{ij}$, which gives the formulas of the abstract; the powers of R follow by dilation.

	entropy	area side	first law at $O(\varepsilon^2)$
electric multipole, degree l	$l+1$	$l+1$	holds
magnetic multipole, degree l	$l+1$	$l+2$	misses by one part in $l+2$

In units of $(c_l/16G)\int_B \psi^2$.

2 The two tidal fields

Conventions for tensors: $(\operatorname{curl} T)_{ij} = \epsilon_{ikl}\partial_k T_{lj}$, $(a \times T)_{ij} = \epsilon_{ikl} a_k T_{lj}$, and sym is the symmetric part.

Electric. A static Newtonian potential Φ of order ε, a harmonic polynomial of degree l:

$$ds^2 = -(1+2\Phi)\,dt^2 + (1-2\Phi)\,d\mathbf{x}^2 + O(\varepsilon^2), \qquad \mathcal{E}_{ij} = \partial_i\partial_j\Phi, \quad \mathcal{B} = 0, \quad \psi_{\mathcal{E}} = l(l-1)\Phi.$$

This is the tide raised by distant masses.

Magnetic. Let φ be a harmonic polynomial of degree l, $v = x \times \nabla\varphi$ and $K = \operatorname{sym}\nabla v$. The perturbation is $h_{ij} = 2\varepsilon t\, K_{ij}$: the slice $t = 0$ is intrinsically flat and has extrinsic curvature εK. (In a stationary gauge the same field is $g_{0i} = -\varepsilon v_i$.) Then

$$\mathcal{E} = 0, \qquad \mathcal{B} = -\varepsilon\, \operatorname{curl} K \quad \text{(static)}, \qquad \psi_{\mathcal{B}} = \tfrac{1}{2}(l-1)l(l+1)\,\varepsilon\varphi.$$

This is the gravitomagnetic tide of distant rotating masses. K is transverse, traceless and harmonic, and $x\cdot K\cdot x = 0$. We write

$$A = \int_B N\,|K|^2, \qquad N = \tfrac{1}{2}(1-r^2).$$

N is the lapse of the conformal Killing vector $\zeta = \frac{1}{2}(1-t^2-r^2)\partial_t - t\,x\cdot\partial$ whose flow maps the diamond to itself. In this notation (3) reads: area side $= \varepsilon^2 A/8G$, entropy $= \frac{l+1}{l+2}\,\varepsilon^2 A/8G$.

3 The entropy side

The observables of the free graviton field are the linearised curvature components smeared with test functions; the algebra of the diamond is generated by those smeared inside D. Three facts give S.

(a) Relative entropy of a coherent state. For a free field, the relative entropy between the vacuum displaced by a classical solution k and the vacuum is [3]

$$S = -\operatorname{Im}\big\langle k,\ P\, i\log\Delta\ k\big\rangle. \tag{4}$$

Here Δ^{it} is the modular group of the diamond and P cuts a solution down to the part that lives in the diamond. In words: flow the classical solution by the modular flow, keep the part of the velocity that lives in the ball, and pair it with the solution itself. For two solutions with data $(h, p = \frac{1}{2}\partial_t h)$, $\operatorname{Im}\langle k, k'\rangle = \frac{1}{32\pi G}\int (h\cdot p' - h'\cdot p)$, half the canonical symplectic form.

(b) The modular flow is geometric. In the vacuum, the modular group of a diamond acts on the curvature as the flow of ζ, with modular parameter 2π times the flow parameter. For the scalar field this is the theorem of Hislop and Longo [4], a conformal image of Bisognano and Wichmann's [5]. For the helicity-2 curvature field we prove it directly, together with Haag duality for the ball (what commutes with everything outside the diamond is exactly the diamond's algebra); see the companion notes [11]. This is the one input of the paper that a reader cannot check from these pages. On the slice the flow reads

$$\delta_\zeta\mathcal{E} = N\,\partial_t\mathcal{E} + 2\operatorname{sym}(x\times\mathcal{B}), \qquad \delta_\zeta\mathcal{B} = N\,\partial_t\mathcal{B} - 2\operatorname{sym}(x\times\mathcal{E}). \tag{5}$$

(c) The computation. For the magnetic multipole the fields are static, so the flow only rotates $\mathcal{B}$ into $\mathcal{E}$, and one line of algebra gives

$$2\operatorname{sym}(x\times\mathcal{B}) = (l+1)\,\varepsilon K.$$

The flowed solution has electric curvature $2\pi(l+1)\varepsilon K$ in the ball. To cut it to the ball we need a metric perturbation with that curvature inside, supported in the ball, and of finite norm. Finite norm means it is continuous across the sphere, so it vanishes there (up to gauge, which does not change the pairing with a transverse K). The profile $2NK$ does this: N vanishes on the sphere,

and $x \cdot K \cdot x = 0$ keeps the constraint satisfied across it. Its linearised Ricci tensor, which is its electric curvature, is

$$\mathrm{Ric}\,(2NK) = (l+2)K,$$

so the cut velocity is $2\pi \cdot \frac{l+1}{l+2} \cdot 2N\varepsilon K$, and

$$S = \frac{2\pi}{32\pi G} \int_B \frac{l+1}{l+2}\, 2N\, \varepsilon K \cdot \varepsilon K = \frac{l+1}{l+2}\, \frac{\varepsilon^2 A}{8G}.$$

The two integers have plain meanings: $l+1$ is the rate at which the modular flow turns $\mathcal{B}$ into $\mathcal{E}$; $l+2$ is the curvature that the lapse profile NK carries.

Second route. Independently, each multipole sector of the graviton can be carried unitarily onto scalar partial waves, local subspaces included (the radial reduction is that of [9]), where Longo and Morsella's formula for the scalar [6] applies [12]. This gives (1) for all l and agrees with the direct route. The two routes share fact (a) and differ in everything else.

Electric data. A duality rotation of $(\mathcal{E}, \mathcal{B})$ is a unitary symmetry of the free field. It maps the vacuum to itself and each local algebra to itself, so it commutes with the modular group. The dual of the magnetic tide is the electric tide with the same ψ (indeed $\mathcal{B} = \frac{1}{2}(l+1)\varepsilon\, \partial\partial\varphi$), and S depends only on the curvature in D. Hence S is the same function of $\psi_{\mathcal{E}}$ as of $\psi_{\mathcal{B}}$, which is (1). The same value follows without duality by solving for the cut velocity directly (checked at $l = 2, 3$).

4 The area side

A piece of pure geometry. For any small change q_{ij} of the flat metric on the ball, Green's identity with $\nabla\nabla N = -\delta_{ij}$ and $N = 0$ on the sphere gives

$$\delta\,\mathrm{Area} - 2\,\delta\,\mathrm{Vol} = -\tfrac{1}{2} \int_B N\, R_{\mathrm{lin}}[q], \tag{6}$$

with R_{lin} the linearised scalar curvature. With the Hamiltonian constraint $R = 16\pi G\, T_{00}$ this is the first law of causal diamonds, $-\delta F/4G = 2\pi \int_B N\, T_{00}$.

Magnetic. The slice $t = 0$ has metric $\delta_{ij} + \varepsilon^2 q_2$ and extrinsic curvature εK. The vacuum Hamiltonian constraint at second order is $R_{\mathrm{lin}}[q_2] = |K|^2$: the magnetic field acts as its own energy density. By (6),

$$F = F_0 - \tfrac{1}{2}\,\varepsilon^2 A, \qquad \text{so} \qquad -\frac{\Delta F}{4G} = \frac{\varepsilon^2 A}{8G},$$

whatever the second-order metric is. It remains to show that the unit ball is still the stationary configuration. It is: the magnetic data do not push the surface at first order (because $K_{nn} = 0$ and $\mathrm{tr}\, K = 0$), the flat-space second variation is non-degenerate except along translations and boosts, and for a single sector parity and time reversal forbid a push along those. A numerical search for the extremum on exact second-order vacuum solutions confirms the value to 5×10^{-10} [11].

Electric. Here the slice itself is curved at first order and the surface moves, so we use an exact solution instead of perturbation theory in a chosen gauge. The static axisymmetric vacuum metrics are Weyl's,

$$ds^2 = -e^{2U}dt^2 + e^{-2U}\big[e^{2\kappa}(d\rho^2 + dz^2) + \rho^2 d\phi^2\big],$$

with U harmonic in flat space and κ given by $\kappa_\rho = \rho(U_\rho^2 - U_z^2)$, $\kappa_z = 2\rho\, U_\rho U_z$. Take $U = \Phi = \varepsilon\, r^l P_l(\cos\theta)$. The slice $t = 0$ is time symmetric, so the stationary surface lies in it. On the surface $r = 1 + \varepsilon a\, P_l$ the expansion to order ε^2 is $F = F_0 + \varepsilon^2 F_2(a)$ with

$$F_2(a) = \frac{2\pi(l-1)(l+2)}{2l+1}\, a^2 - \frac{8\pi(l-1)}{2l+1}\, a + \frac{12\pi(l-1)}{(2l+1)(2l+3)},$$

computed in exact arithmetic by two independent codes (full expansion for $l \le 6$, a reduced expansion for $l \le 12$). It is stationary at $a = 2/(l+2)$, where

$$-2F_2 = \frac{8\pi\, l(l-1)}{(l+2)(2l+1)(2l+3)} = \frac{4\pi}{35},\ \frac{16\pi}{105},\ \frac{16\pi}{99},\ \frac{160\pi}{1001},\ \frac{2\pi}{13},\ \dots$$

in units $\varepsilon^2/8G$. This equals (1) identically in l. A stationary value of F does not depend on coordinates, so no gauge choice enters. The script also checks (6), checks that the slice is scalar-flat, and fails when any one ingredient is altered. For $l = 1$ it returns $F_2 \equiv 0$: a uniform field is flat through first order, and its second-order vacuum curvature cannot change F at this order, by (6). Other m at the same l follow because both sides are rotation-invariant quadratic forms on the irreducible degree-l multiplet, hence multiples of $\int \psi^2$; one m fixes the multiple.

5 What the mismatch is not

Not removable by a scalar. Every parity-even quadratic Lorentz scalar built from the curvature changes sign under the duality map $\mathcal{E} \to \mathcal{B}$, $\mathcal{B} \to -\mathcal{E}$; for $l = 2$ it is $R_{abcd}R^{abcd} \propto \mathcal{E}^2 - \mathcal{B}^2$. So a local scalar correction cannot vanish on electric data and be non-zero on magnetic data. It can move the mismatch between the sectors but not remove it: for $l = 2$,

$$\text{area side} = \tfrac{7}{6}\, S - \frac{\pi R^6}{2520\, G}\big(\mathcal{E}_{ij}\mathcal{E}_{ij} - \mathcal{B}_{ij}\mathcal{B}_{ij}\big),$$

and $S \propto \mathcal{E}^2 + \mathcal{B}^2$ is not a scalar. Jacobson's proposal for non-conformal matter is that the modular energy differs from the stress-energy flux by a local scalar, which the cosmological-constant term absorbs [2]; Casini, Galante and Myers showed where that fails [7]. A scalar gives proportionality here at best, never equality.

Not the scalar-field effect. For comparison, take a minimally coupled massless scalar with a static harmonic profile φ. By (6) the area side is $2\pi\cdot\frac{1}{2}\int N|\nabla\varphi|^2$, while the entropy is $2\pi\big(\frac{1}{2}\int N|\nabla\varphi|^2 + \frac{1}{2}\int\varphi^2\big)$ [6]. There the entropy *exceeds* the area side, by the integral of a scalar density. For gravitons the entropy falls *short*. (For a conformally coupled scalar the Wald term $-\frac{\pi}{3}\oint\varphi^2$ restores the balance.)

Not an edge-mode ambiguity. Any algebra one assigns to the ball that (i) is related to the full theory by a fixed quantum channel, (ii) cannot signal to the outside, and (iii) contains the interior curvature observables, has the same relative entropy [11]. Centres and edge modes of this kind do not move S.

What is already in print. For $l = 2$ the area side is known. Eq. (75) of Jacobson, Senovilla and Speranza [8] gives the second-order area deficit of a small ball at fixed volume in vacuum, for a one-parameter family of shapes. At the constant-mean-curvature shape it reads $\delta A = -\frac{\pi \ell^6}{105}\left(\mathcal{E}^2 + \frac{4}{3}\mathcal{B}^2\right)$ in four dimensions, which is $-4G$ times the area terms above. Two things follow from their formula:

- No shape makes the deficit proportional to the Bel–Robinson super-energy $W = \mathcal{E}^2 + \mathcal{B}^2$.
- The magnetic coefficient is the same for every shape. The electric one is not: it equals the entropy only at the stationary shape. For a geodesic ball of fixed volume, the construction of [2], it is $\mathcal{E}^2/6$ in place of $3\mathcal{E}^2/4$, and the electric case fails too.

What we did not find in print is the entropy side, $S = \pi R^6 W/420G$, and hence the comparison. (Benedetti and Casini [9] computed the vacuum entanglement entropy of gravitons in a sphere, a different quantity.) These are search results, not a priority claim.

6 Where the extra 1 comes from

The two integers $l+1$ and $l+2$ are two rates: how fast the modular flow moves the curvature, and how fast the conformal Killing flow moves the metric. For any lapse function N and any transverse, traceless, harmonic K,

$$\underbrace{\mathrm{Ric}\,(2NK)}_{\text{geometric rate}} - \underbrace{2\,\mathrm{sym}(\nabla N \times \mathrm{curl}\,K)}_{\text{modular rate}} = 2\,\mathrm{sym}(K\cdot\nabla\nabla N) - (\nabla^2 N)\,K. \tag{7}$$

For the Rindler wedge $N = x_1$ is the lapse of a boost, a true symmetry, $\nabla\nabla N = 0$, and the two rates are equal: no mismatch, as must be the case on a Killing horizon. For the ball $\nabla\nabla N = -\delta_{ij}$ and the right side is exactly K. The generator that moves the metric and the generator that moves the curvature differ by exactly the non-Killing term. In line with this, $(l+1)/(l+2) \to 1$ as $l \to \infty$, where the field hugs the sphere and the diamond looks locally like a wedge.

7 How it was checked

Claim	Support
Entropy formula (1)	Two proofs with different ingredients; they agree for every l. Both rest on fact (b).
Fact (b) and duality for the ball	Proved in the companion notes [11]. Not checked by anyone outside this programme.
Entropy, numerically	A three-dimensional lattice of the graviton field with no multipole reduction. For the uniform magnetic field the extrapolated ratio to (1) is 1.00, bracket 0.89 to 1.02; if the entropy equalled the area side it would be 1.33. For electric data of $l = 2, 3, 4$ the ratios are 1.02, 1.01, 0.99.
Area side, magnetic	Identity (6) and the constraint; exact nonlinear solutions to 5×10^{-10}; for $l = 2$, eq. (75) of [8].
Area side, electric	Exact Weyl solution, $l \le 12$; for $l = 2$, eq. (75) of [8].
Killing limit	Identity (7) on random data.

All checks so far are internal to this programme. No outside referee has read the proofs.

8 Assumptions, and what would refute the result

1. **Scope.** Free linearised gravity at order G^0, four dimensions, a flat ball, one multipole order at a time, static electric or stationary magnetic data. Whether the ball algebra of full quantum gravity reduces to this free algebra as $G \to 0$ is not claimed. Constructions that add an observer or a clock [10] may change the answer, and this is the assumption we consider most open.

2. **The area side.** That the gravitational term of the first law at this order is $-\Delta F/4G$ at the stationary configuration. This is the charge the first law gives for matter [1]; at this order it equals the area deficit at fixed volume of the constant-mean-curvature surface. The choice of surface matters for electric data and not for magnetic data (Section 5). Whether the construction of [10] yields exactly this term at second order we have not established.

3. **The algebra.** The three conditions (i)–(iii) above.

4. **Einstein's equations** hold through second order. Only the Hamiltonian constraint on the slice is used.

The result would be refuted by: an error in the proof that the helicity-2 modular flow is (5); a second-order first law in print that balances on magnetic data; or a demonstration that the stationary surface for magnetic data is not the one used here. Mixtures of different l, and mixtures of electric and magnetic data with a non-zero super-Poynting vector, push the ball at second order and are not covered.

9 Questions

1. The magnetic area coefficient is the same for every shape of ball at fixed volume, and the entropy coefficient is three quarters of it. Is there a graviton analogue of the Wald term, which restores the balance for a conformally coupled scalar, that would do the same here?

2. Is the electric agreement at the stationary shape significant, or an accident of one shape?

3. Is a mismatch of this kind known, in this or another language?

Changes since version 1

A checker who saw only the paper recomputed its formulas independently, with instructions to break it. Every formula and number agreed. Four statements were wrong or too broad and are corrected: the area formula had been stated for mixed electric and magnetic data without a condition; the $l = 1$ control had been called flat space; the first law had been said to hold for matter, not conformal matter; and reference [8] had been described as a different prescription, when its eq. (75) contains the $l = 2$ area terms. The claim that the area side was new at $l = 2$ is withdrawn. Version 3 only shortens the text.

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